

Transportation Telemetry Predictive Timing Platform

Synthetic train telemetry, lakehouse pipeline, and neural network delay prediction

Project question

Can a real-time transportation telemetry platform predict train delay timing and support predictive maintenance decision-making using sensor, location, and operational event data?

Overview Metrics Figures **Architecture**

Architecture

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Train telemetry sensors
    ↓
Kafka / Kinesis concept
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Spark / Flink concept
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Bronze: raw immutable events
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Silver: cleansed telemetry records
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Gold: ML-ready certified feature table

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Neural network predictive timing model

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Dashboards and operations analytics

Reliability concepts

- event time vs processing time
- late-arriving events
- checkpointing
- replayable topics
- deduplication
- schema evolution
- SLA monitoring